

Stochastic reachability-integrated flexible formation control for multi-agent systems

Yutong Zhu ^{a,b}, Ye Zhang ^{b,*}

^a ICTEAM Institute, University of Louvain, Louvain-la-Neuve, 1348, Belgium

^b School of Astronautics, Northwestern Polytechnical University, Xi'an, 710072, China

ARTICLE INFO

Keywords:

Multi-agent systems
Control barrier function
Safety-critical unmanned autonomous systems
Probabilistic reachability
Flexible formation control

ABSTRACT

This paper introduces a unified framework for flexible formation safety control in multi-agent systems (MAS), specifically tailored for safety-critical unmanned autonomous systems. Existing rigid formation control methods often lack the flexibility to negotiate complex, clutter environments, while traditional safety filters can be overly conservative or computationally prohibitive. To address these challenges, this paper introduces a unified framework enabling formations to deform temporarily for obstacle 'pass-through' while maintaining consensus and trajectory tracking. The method integrates distributed nonlinear model predictive control (NMPC) for consensus-driven optimization, control barrier functions (CBFs) for real-time safety enforcement in safety-critical environments, and probabilistic reachability analysis to handle uncertainties with bounded violation probabilities. Key innovations include a novel stochastic control barrier-value function (CBVF) derived from Hamilton-Jacobi-Isaacs variational inequalities (HJI-VIs), which computes maximal, deformable safe sets via scenario-based approximations to ensure distributed safety control, and a scenario-sampled distributed NMPC ensuring recursive feasibility under stochastic disturbances. Theoretical proofs establish asymptotic stability of formation errors, forward invariance of safe sets, and probabilistic safety bounds under bounded disturbances in safety-critical scenarios. Simulations on drones, as representative unmanned autonomous systems, demonstrate the framework's efficacy, where formations adapt to traversing obstacles with <5% violation probability under Gaussian noise, outperforming baseline methods in flexibility, computational efficiency, and resilience for distributed safety control.

1. Introduction

Multi-agent systems (MAS) have revolutionized applications in robotics, aerospace, and autonomous transportation, enabling coordinated behaviors that surpass the capabilities of individual agents [1–3]. Formation control, a cornerstone of MAS coordination, involves maintaining predefined geometric configurations among agents while tracking desired trajectories or achieving collective objectives [4,5]. In safety-critical scenarios involving unmanned autonomous systems, such as urban air mobility, search-and-rescue operations, or warehouse logistics, formations must adapt to dynamic environments riddled with uncertainties, including external disturbances, model inaccuracies, and unknown obstacles [6,7]. Traditional rigid formation control paradigms, which enforce strict inter-agent distances, often fail in such safety-critical settings, leading to mission failure, excessive conservatism, or unsafe behaviors [8,9].

* Corresponding author.

E-mail address: yutong.zhu@mail.nwpu.edu.cn, zhang_ye@nwpu.edu.cn (Y. Zhang).

<https://doi.org/10.1016/j.jfranklin.2026.108578>

Received 13 October 2025; Received in revised form 2 March 2026; Accepted 4 March 2026

Available online 6 March 2026

0016-0032/© 2026 The Franklin Institute. Published by Elsevier Inc. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

A key emerging requirement is flexible formation control for safety-critical unmanned autonomous systems, where agents can temporarily deform their configuration to accommodate environmental perturbations without compromising overall mission goals [4,10]. This flexibility is particularly vital for obstacle avoidance in distributed safety control: rather than halting or rerouting the entire formation upon encountering an obstacle, agents should allow the obstacle to "pass through" the formation, reforming seamlessly afterward. Such behavior mimics natural swarms, like schools of fish or flocks of birds, which exhibit emergent resilience to intrusions [11–13]. However, achieving this in engineered systems is challenging, especially when obstacles are unknown (e.g., detected only via onboard sensors) and the formation must remain safe under stochastic disturbances in safety-critical environments [14,15]. Existing methods to formation control and safety can be broadly categorized into optimization-based, Lyapunov-like, and reachability-driven methods. Optimization-based techniques, such as MPC, excel in handling constraints and predictions but often struggle with nonlinearity and distribution in real-time [16,17]. Distributed MPC variants, including nonlinear MPC (NMPC), enable scalable coordination via consensus mechanisms, as seen in wheeled mobile robot formations [18]. However, they typically assume known environments and may not guarantee safety under uncertainty without additional safeguards for safety-critical unmanned autonomous systems.

Control barrier functions (CBFs) provide a powerful Lyapunov-inspired tool for enforcing safety constraints in real-time, rendering sets forward-invariant through quadratic programming (QP) filters [19–21]. CBFs have been extended to MAS for collision avoidance [22], but constructing valid CBFs for complex dynamics remains ad hoc, often leading to conservative safe sets that hinder flexibility [20]. Moreover, standard CBFs do not inherently handle probabilistic uncertainties or unknown obstacles, limiting their applicability in stochastic safety-critical environments. Reachability analysis, rooted in Hamilton-Jacobi (HJ) theory, offers formal safety guarantees by computing reachable sets (regions agents can reach or avoid under worst-case) or probabilistic disturbances [23,24]. Probabilistic reachability, in particular, reduces conservatism by bounding violation probabilities using stochastic HJ equations and scenario programs [23]. Recent integrations of reachability with CBFs, such as control barrier-value functions (CBVFs), unify viscosity solutions to HJ variational inequalities with barrier constraints, enabling maximal safe sets [25]. Yet, these methods are seldom applied to distributed formations in unmanned autonomous systems, where scalability and flexibility are paramount for distributed safety control. In noncooperative or adversarial settings, such as pursuit-evasion (PE) games, formation control must incorporate herding or containment strategies [26–28]. For instance, indirect herding frameworks decouple seeking and herding phases, using backstepping for trajectory tracking and potential functions for collision avoidance [29]. These inspire flexible formations but lack probabilistic guarantees and distributed optimization for unknown obstacles in safety-critical scenarios. Despite these advances, critical gaps persist: i) rigid formations are brittle to unknown obstacles, disallowing "pass-through" behaviors; ii) distributed methods often overlook probabilistic safety under uncertainty in unmanned autonomous systems; and iii) integrations of NMPC, CBFs, and reachability are rare, missing opportunities for synergistic benefits like predictive flexibility, real-time enforcement, and formal guarantees for distributed safety control.

Unlike traditional rigid MPC approaches [30,31] that enforce strict geometric constraints, often leading to infeasibility in cluttered environments, the proposed framework enables flexible formation deformation. Furthermore, distinct from standard deterministic CBFs [20] that rely on worst-case assumptions yielding overly conservative safe sets, our stochastic control barrier-value function (CBVF) explicitly incorporates probabilistic reachability. This allows the system to quantify safety risks (bounded by probability) and dynamically trade off between safety conservatism and formation flexibility, a feature absent in existing rigid or deterministic safety filters. Based on current research [25,32] and our prior work [4,29,33], this paper addresses these gaps by proposing a unified framework for flexible formation safety control in MAS with one leader and two followers, emphasizing distributed safety control of unmanned autonomous systems. The leader tracks a reference trajectory, while followers maintain consensus-based offsets that adapt dynamically. Obstacles, assumed unknown and detected online, can traverse the formation without collision, enabled by deformable safe sets. The framework integrates:

1. Distributed NMPC for consensus-driven trajectory optimization, ensuring formation maintenance while predicting flexible deformations in safety-critical environments.
2. CBFs for real-time safety filters, augmented with adaptive parameters to allow obstacle intrusion without violating inter-agent constraints in unmanned autonomous systems.
3. Probabilistic Reachability Analysis to compute stochastic reachable sets, providing bounded probability safety guarantees under uncertainties like wind gusts or sensor noise.

Key innovations include a novel CBF-reachability hybrid that solves a stochastic Hamilton-Jacobi-Isaacs variational inequality (HJI-VI) for flexible safe sets, and a distributed NMPC formulation with scenario-based sampling for recursive feasibility. Rigorous proofs establish asymptotic stability, forward invariance, and probabilistic bounds on constraint violations. The contributions are threefold:

1. **Stochastic CBVF construction:** we develop a stochastic CBVF that merges CBFs with probabilistic reachability via HJI-VIs, enabling maximal, deformable safe sets that adapt to unknown obstacles while bounding safety violation probabilities. This addresses conservatism in traditional CBFs and extends deterministic CBVFs to stochastic settings with scenario approximations for tractability in safety-critical unmanned autonomous systems.
2. **Scenario-based distributed NMPC:** we propose a distributed NMPC augmented with scenario sampling to incorporate probabilistic constraints, ensuring recursive feasibility and consensus under disturbances. This integration allows flexible deformations without centralized computation, with proofs of asymptotic formation stability for distributed safety control.

3. Theoretical and empirical validation: we provide rigorous guarantees on forward invariance, probabilistic safety (with violation bounds $\leq \delta$), and input-to-state stability, complemented by drone simulations demonstrating real-time performance, reduced conservatism, and resilience to traversing obstacles in uncertain environments for unmanned autonomous systems.

The structure of the paper is organized as follows. [Section 2](#) formulates the system dynamics and problem. [Section 3](#) presents overall framework on distributed NMPC, CBFs, and reachability. [Section 4](#) details the proposed control method. [Section 5](#) provides simulation results. Main conclusions with future directions are drawn in [Section 6](#).

2. System model and problem formulation

2.1. Agent dynamics

Consider a multi-agent system consisting of one leader (indexed as $i = 0$) and two followers ($i = 1, 2$), each modeled as a nonlinear control-affine system subject to bounded disturbances:

$$\dot{x}_i(t) = f(x_i(t)) + g(x_i(t))u_i(t) + d_i(t), \quad i = 0, 1, 2 \quad (1)$$

where $x_i(t) \in \mathcal{X} \subseteq \mathbb{R}^{n_x}$ is the state of agent i , $u_i(t) \in \mathcal{U} \subseteq \mathbb{R}^{n_u}$ is the control input, and $d_i(t) \in \mathcal{D} \subseteq \mathbb{R}^{n_d}$ is a bounded disturbance representing uncertainties such as wind gusts or model errors. The functions $f : \mathbb{R}^{n_x} \rightarrow \mathbb{R}^{n_x}$ and $g : \mathbb{R}^{n_x} \rightarrow \mathbb{R}^{n_x \times n_u}$ are assumed to be locally Lipschitz continuous in x_i . The state space $\mathcal{X}$ is path-connected and contains the origin, while $\mathcal{U}$ and $\mathcal{D}$ are compact convex sets.

For concreteness, we adopt a quadrotor UAV model as in Ref[29], where $n_x = 12$, $x_i = [p_i^\top, v_i^\top, \Phi_i^\top, \omega_i^\top]^\top$ with position $p_i \in \mathbb{R}^3$, velocity $v_i \in \mathbb{R}^3$, Euler angles $\Phi_i = [\phi_i, \theta_i, \psi_i]^\top \in \mathbb{R}^3$, and angular velocity $\omega_i \in \mathbb{R}^3$. The dynamics are:

$$\begin{aligned} \dot{p}_i &= v_i, \quad \dot{v}_i = -ge_3 + \frac{1}{m_i} R(\Phi_i)T_i + d_{v,i} \\ \dot{\Phi}_i &= W(\Phi_i)\omega_i, \quad \dot{\omega}_i = J_i^{-1}(\tau_i - \omega_i \times J_i\omega_i) + d_{\omega,i} \end{aligned} \quad (2)$$

where $m_i > 0$ is the mass, $J_i > 0$ is the inertia matrix, g is gravity, $e_3 = [0, 0, 1]^\top$, $R(\Phi_i)$ is the rotation matrix, $W(\Phi_i)$ is the angular velocity transformation matrix, $T_i = [0, 0, u_{T,i}]^\top$ is the thrust vector with $u_{T,i} \in [u_{T,\min}, u_{T,\max}]$, and $\tau_i \in \mathbb{R}^3$ is the torque vector with bounds. The disturbances $d_{v,i}, d_{\omega,i}$ satisfy $|d_{v,i}| \leq \bar{d}_v$ and $|d_{\omega,i}| \leq \bar{d}_\omega$. The leader ($i = 0$) tracks a reference trajectory $x_r(t)$ generated by a higher-level planner, while followers maintain flexible offsets relative to the leader.

2.2. Flexible formation configuration

The formation is defined by time-varying nominal offsets $\Delta_i(t) \in \mathbb{R}^3$ for followers relative to the leader's position $p_0(t)$: $p_i^d(t) = p_0(t) + \Delta_i(t)$, where $\Delta_i(t)$ is designed to allow deformation for obstacle passage. Flexibility is achieved by relaxing rigid distance constraints, permitting $|p_i(t) - p_j(t)| \in [r_{sep}^{\min}, r_{sep}^{\max}]$ with $r_{sep}^{\min} > 0$ to avoid collisions and $r_{sep}^{\max}$ to prevent dispersion.

Communication is modeled by a directed graph $\mathcal{G} = (\mathcal{V}, \mathcal{E}, A)$, where $\mathcal{V} = \{0, 1, 2\}$, $\mathcal{E} \subseteq \mathcal{V} \times \mathcal{V}$, and $A = [a_{ij}] \in \mathbb{R}^{3 \times 3}$ is the weighted adjacency matrix with $a_{ij} > 0$ if $(j, i) \in \mathcal{E}$ (information flow from j to i). The graph is assumed strongly connected with the leader as root. The in-degree matrix $D = \text{diag}(\sum_j a_{ij})$, Laplacian $L = D - A$, and leader adjacency $b = [b_i]^\top$ with $b_0 = 0$, $b_i > 0$ if follower i receives leader information. The consensus-based formation error for follower i is:

$$e_i(t) = \sum_{j=1}^2 a_{ij}(p_i(t) - p_j(t) - \Delta_{ij}(t)) + b_i(p_i(t) - p_0(t) - \Delta_i(t)) \quad (3)$$

where $\Delta_{ij}(t) = \Delta_i(t) - \Delta_j(t)$. The objective is $\lim_{t \rightarrow \infty} e_i(t) = 0$ while allowing transient deformation for obstacles.

2.3. Obstacle model

Obstacles are assumed unknown a priori and detected online via onboard sensors with detection range $R_s > d_{\max}$. Each obstacle o_j is modeled as a sphere with center $c_j(t)$ and radius $R_j(t)$, potentially moving with bounded velocity $|\dot{c}_j| \leq \bar{v}_o$. The safe set for agent i w.r.t. o_j is $C_{ij} = x_i : |p_i - c_j| \geq R_i + R_j + \epsilon_s$, where $\epsilon_s > 0$ is a safety margin and R_i is the agent's bounding radius. Probabilistic reachability ensures $P(x_i(t) \in \bigcap_j C_{ij}, \forall t) \geq 1 - \delta$ with risk bound $\delta > 0$.

2.4. Problem statement

The objective is to design distributed controllers $u_i(t)$ such that:

- 1) Formation Maintenance: Followers achieve $\lim_{t \rightarrow \infty} e_i(t) = 0$ while allowing flexible deformation for obstacle passage.
- 2) Trajectory Tracking: The leader tracks $x_r(t)$, i.e., $\lim_{t \rightarrow \infty} |x_0(t) - x_r(t)| = 0$.
- 3) Safety Guarantees: With probability $\geq 1 - \delta$, the safe set for agent i is defined with respect to two types of entities: obstacles (indexed by $j \in \{1, \dots, M\}$) and other agents (indexed by $l \in \{1, \dots, N\} \setminus \{i\}$). Agents avoid collisions: $x_i(t) \in \bigcap_{j=1}^M C_{ij} \cap \bigcap_{l \neq i} C_{il}$, $\forall t$, where C_{ij} denotes safety with respect to obstacle j , and C_{il} denotes safety with respect to agent l , and $C_{il} = x_i : |p_i - p_l| \geq 2R_i + \epsilon_s$.
- 4) Recursive Feasibility: The NMPC problems remain feasible under disturbances.

Subject to state/input constraints $x_i(t) \in \mathcal{X}$, $u_i(t) \in \mathcal{U}$, and communication graph $\mathcal{G}$. The framework integrates distributed NMPC for optimization, CBFs for real-time safety, and probabilistic reachability for uncertainty handling, with proofs of stability and feasibility.

3. Overall framework

This section establishes the foundational concepts underpinning the proposed framework. We review distributed nonlinear model predictive control (NMPC) for consensus-based optimization in multi-agent systems, control barrier functions (CBFs) for real-time safety enforcement, and probabilistic reachability analysis for handling uncertainties. These tools are adapted to the nonlinear, control-affine dynamics of [Section 2](#).

3.1. Distributed nonlinear model predictive control

Distributed NMPC enables scalable trajectory optimization in MAS by allowing each agent to solve a local optimal control problem while exchanging predictions with neighbors to achieve consensus. For agent $i \in \{0, 1, 2\}$, consider the discrete-time approximation of the dynamics in (1) over a prediction horizon N_p , obtained via Euler discretization with sampling time Δt :

$$x_i(k+1) = x_i(k) + \Delta t(f(x_i(k)) + g(x_i(k))u_i(k) + d_i(k)) \quad (4)$$

where k denotes the discrete time step, and $d_i(k) \in \mathcal{D}$ is treated as a predicted disturbance (nominal $d_i(k) = 0$ for deterministic planning, with robustness via reachability in the following section). The local NMPC problem for agent i at time k is formulated as:

$$\min_{\mathbf{u}_i} \sum_{\ell=0}^{N_p-1} \ell_i(x_i(k+\ell|k), u_i(k+\ell|k)) + \ell_{f,i}(x_i(k+N_p|k)) \quad (5)$$

subject to

$$\begin{aligned} x_i(k+\ell+1|k) &= x_i(k+\ell|k) + \Delta t(f(x_i(k+\ell|k)) + g(x_i(k+\ell|k))u_i(k+\ell|k)), \quad \ell = 0, \dots, N_p-1 \\ x_i(k|k) &= x_i(k), \quad x_i(k+\ell|k) \in \mathcal{X}, \quad u_i(k+\ell|k) \in \mathcal{U} \\ \|p_i(k+\ell|k) - p_j(k+\ell|k) - \Delta_{ij}(k+\ell|k)\| &\leq \epsilon_c, \quad \forall j \in \mathcal{N}_i \end{aligned} \quad (6)$$

where $\mathbf{u}_i = [u_i(k|k)^\top, \dots, u_i(k+N_p-1|k)^\top]^\top$ is the predicted input sequence, $x_i(k+\ell|k)$ denotes the state predicted at step k for time $k+\ell$, $\ell_i(\cdot, \cdot)$ is a stage cost (e.g., quadratic in tracking error and control effort), $\ell_{f,i}(\cdot)$ is a terminal cost, $\mathcal{N}_i = j : (j, i) \in \mathcal{E}$ is the neighbor set, and $\epsilon_c > 0$ is a consensus tolerance allowing flexible deformation. For the leader ($i = 0$), the stage cost includes reference tracking:

$$\ell_0 = \|x_0 - x_r\|_Q^2 + \|u_0\|_R^2 \quad (7)$$

with positive definite Q, R . Followers incorporate consensus terms:

$$\ell_i = \|e_i\|_Q^2 + \|u_i\|_R^2 + \sum_{j \in \mathcal{N}_i} a_{ij} \|p_i - p_j - \Delta_{ij}\|_S^2 + b_i \|p_i - p_0 - \Delta_i\|_S^2 \quad (8)$$

where $S > 0$.

To ensure distributed solvability, agents exchange predicted trajectories $x_j(k+\ell|k)_{\ell=0}^{N_p}$ over $\mathcal{G}$, approximating neighbors' predictions in the consensus constraints. Recursive feasibility is guaranteed under assumptions of stabilizable dynamics and Lipschitz continuity of f, g . The first element of the optimal $\mathbf{u}_i$ is applied: $u_i(k) = u_i(k|k)^\top$, with re-optimization at $k+1$.

Assumption 1. The graph $\mathcal{G}$ is strongly connected, and the cost functions are continuously differentiable with bounded Hessians.

Under [Assumption 1](#), the distributed NMPC achieves asymptotic consensus: $\lim_{k \rightarrow \infty} e_i(k) = 0$ if undisturbed, as proven via Lyapunov descent on the collective cost.

3.2. Control barrier function

CBFs enforce forward invariance of safe sets in real-time, complementing NMPC's predictive capabilities. For a continuously differentiable function $h_i : \mathbb{R}^{n_x} \rightarrow \mathbb{R}$, the safe set is defined as: $C_i = \{x_i \in \mathbb{R}^{n_x} \mid h_i(x_i) \geq 0\}$. This set is rendered forward-invariant if there exists a locally Lipschitz extended $\mathcal{K}_\infty$ -class function $\alpha : \mathbb{R} \rightarrow \mathbb{R}$ such that for all $x_i \in C_i$:

$$\sup_{u_i \in \mathcal{U}} \left[\frac{\partial h_i}{\partial x_i} (f(x_i) + g(x_i)u_i + d_i) + \alpha(h_i(x_i)) \right] \geq 0, \quad \forall d_i \in \mathcal{D} \quad (9)$$

where h_i is a robust CBF, ensuring C_i is forward-invariant under bounded disturbances (i.e., if $x_i(0) \in C_i$, then $x_i(t) \in C_i$ for all $t \geq 0$). Intuitively, forward invariance implies that if the system starts in a safe state, it will remain in the safe state for all future time, effectively creating an inescapable 'safe bubble' around the agent [\[34\]](#).

For inter-agent collision avoidance, define $h_{il}(x_i, x_l) = \|p_i - p_l\|^2 - (2R_i + \epsilon_s)^2 \geq 0$ for $l \neq i$. For obstacles, upon detection of o_j , set $h_{ij}(x_i, c_j) = \|p_i - c_j\|^2 - (R_i + R_j + \epsilon_s)^2 \geq 0$. The composite CBF for agent i is $h_i = \min_{l \neq i, j} \{h_{il}, h_{ij}\}$, allowing flexible "pass-through" by temporarily relaxing non-essential h_{il} via adaptive α (e.g., $\alpha(r) = \gamma r$ with $\gamma > 0$ tunable based on deformation needs). The NMPC-generated nominal control $u_{i,\text{nom}}$ is filtered through a QP to ensure safety:

$$u_i^* = \arg \min_{u_i \in \mathcal{U}} \|u_i - u_{i,\text{nom}}\|^2 \quad (10)$$

subject to

$$\frac{\partial h_i}{\partial x_i} (f(x_i) + g(x_i)u_i) + \inf_{d_i \in D} \frac{\partial h_i}{\partial x_i} d_i + \alpha(h_i) \geq 0 \quad (11)$$

The infimum over D is computed explicitly for polytopic D (e.g., via vertex enumeration), ensuring robustness. Under Lipschitz continuity of h_i and controllability assumptions, the QP is feasible if $x_i \in \text{int}(C_i)$.

Lemma 1. If h_i is a robust CBF satisfying (9) and u_i^* solves the QP (10)-(11), then the safe set $C_i = \{x_i : h_i(x_i) \geq 0\}$ is forward-invariant for the closed-loop system under bounded disturbances $d_i \in D$.

Proof. The proof proceeds in two steps: (i) establishing that the CBF condition ensures non-decrease along the boundary, and (ii) integrating to show no escape from C_i .

Assume $x_i(0) \in C_i$, so $h_i(x_i(0)) \geq 0$. By definition of a robust CBF, for all $x_i \in C_i$, there exists $u_i \in \mathcal{U}$ such that

$$\sup_{u_i \in \mathcal{U}} \left[\frac{\partial h_i}{\partial x_i} (f(x_i) + g(x_i)u_i) + \inf_{d_i \in D} \frac{\partial h_i}{\partial x_i} d_i + \alpha(h_i(x_i)) \right] \geq 0 \quad (12)$$

where $\alpha \in \mathcal{K}_\infty$ extended (locally Lipschitz, $\alpha(0) = 0$, strictly increasing). The QP (10)-(11) minimizes deviation from the nominal $u_{i,\text{nom}}$ while enforcing this inequality, ensuring feasibility for $x_i \in \text{int}(C_i)$ under relative degree 1 and controllability assumptions [35]. Under the applied u_i^* , the Lie derivative satisfies

$$\mathcal{L}_f h_i + \mathcal{L}_g h_i u_i^* + \inf_{d_i} \mathcal{L}_d h_i + \alpha(h_i) \geq 0 \quad \forall x_i \in \partial C_i \quad (13)$$

where $\mathcal{L}_d h_i = \frac{\partial h_i}{\partial x_i} d_i$. Thus, on the boundary ($h_i = 0$),

$$\dot{h}_i \geq -\alpha(h_i) + \inf_{d_i} \mathcal{L}_d h_i + \mathcal{L}_g h_i u_i^* \geq 0 \quad (14)$$

preventing outward flow. Integrating, consider the comparison function $\phi(t)$ solving $\dot{\phi} = -\alpha(\phi)$, $\phi(0) = h_i(0) \geq 0$. By the comparison lemma [36], $h_i(t) \geq \phi(t) \geq 0$ for all $t \geq 0$, as $\alpha(r) > 0$ for $r > 0$ ensures $\phi(t) > 0$ if $\phi(0) > 0$, and $\phi(t) = 0$ if $\phi(0) = 0$. Bounded disturbances are handled via the infimum, yielding robust forward invariance. $\square$

3.3. Probabilistic reachability analysis

Probabilistic reachability quantifies safety under stochastic disturbances by computing reachable sets with bounded violation probabilities, reducing conservatism compared to deterministic methods. Model $d_i(t)$ as a Wiener process: $d_i(t) = \sigma(x_i, t)dW(t)$, where $W(t)$ is standard Brownian motion and σ is Lipschitz. The probabilistic forward reachable set (PRS) from initial set $\mathcal{K}_i$ over $[t_0, t_f]$ with probability level $\rho \in (0, 1)$ is:

$$\overline{\text{PRS}}_\rho(t_0, t_f; \mathcal{K}_i) = \{x_i(t_f) : P(x_i(t) \in \mathcal{X}_{\text{safe}}, \forall t \in [t_0, t_f] \mid x_i(t_0) \in \mathcal{K}_i) \geq \rho\} \quad (15)$$

where $\mathcal{X}_{\text{safe}} = \bigcap_{l,j} C_{il} \cap C_{ij}$. Similarly, the backward reachable set $\overline{\text{PRS}}_\rho$ defines states reaching a target with probability $\geq \rho$. PRS computation solves a stochastic Hamilton-Jacobi-Bellman (HJB) equation for value function $V_i(x_i, t)$:

$$\frac{\partial V_i}{\partial t} + \min_{u_i} \left\{ \frac{\partial V_i}{\partial x_i} (f(x_i) + g(x_i)u_i) + \frac{1}{2} \text{tr} \left(\sigma^\top \frac{\partial^2 V_i}{\partial x_i^2} \sigma \right) \right\} = 0 \quad (16)$$

with terminal $V_i(x_i, t_f) = l_i(x_i)$, where $l_i(x_i) = \min\{h_i(x_i), 0\}$ encodes safety.

The level set $x_i : V_i(x_i, t) \geq \log \rho$ approximates $\overline{\text{PRS}}_\rho$. For scalability, use scenario-based approximation: Sample N_s disturbance realizations $\{d_i^{(m)}(t)\}_{m=1}^{N_s}$, solve deterministic optimizations, and bound suboptimality via support rank.

Lemma 2. For confidence $\beta \in (0, 1)$ and violation tolerance $\epsilon > 0$, select $N_s \geq \frac{2}{\epsilon} (\ln \frac{1}{\beta} + n_x - 1)$. The scenario-approximated value function V_i^* satisfies $\mathbb{P}(V_i^* \leq V_i^{\text{opt}} + \epsilon) \geq 1 - \beta$, ensuring PRS violation probability $\leq \delta = 1 - \rho + \beta$, where ρ is the target probability level.

Proof. This follows scenario optimization theory [37]. Sample N_s independent disturbance realizations $\{d_i^{(m)}(t)\}_{m=1}^{N_s} \sim \mu$ (stochastic measure). Solve the deterministic program approximating (16):

$$\begin{aligned} V_i^*(x_i, t) &= \min_m l_i(x_i^{(m)}(t_f)) \\ \text{s.t. } \dot{x}_i^{(m)} &= f(x_i^{(m)}) + g(x_i^{(m)})u_i + d_i^{(m)} \end{aligned} \quad (17)$$

with terminal $l_i(x_i) = \min\{h_i(x_i), 0\}$. The support rank of the program is at most n_x (dimension of decision variables). By the one-sided Chernoff bound and compression, the probability that a new sample violates $V_i^* \leq V_i^{\text{opt}} + \epsilon$ is $\leq \epsilon$ with confidence $1 - \beta$. Integrating over μ , the PRS $\overline{\text{PRS}}_\rho = \{x_i(t_f) : V_i(x_i, t) \geq \log \rho\}$ has violation $\mathbb{P}(x_i(t) \notin \mathcal{X}_{\text{safe}}) \leq 1 - \rho + \beta = \delta$, as the scenario solution is ϵ -suboptimal in expectation. $\square$

These preliminaries enable the hybrid framework in the next section, where NMPC optimizes trajectories within PRS-bounded tubes, filtered by CBF-QPs for safety.

The stochastic CBVF is posited as

$$B_i^\gamma(x_i, t) = \min\{V_i(x_i, t), \gamma h_i(x_i)\} \quad (22)$$

reconciling the reachability value V_i (ensuring viability with $V_i \geq 0$) and the scaled barrier γh_i . By the dynamic programming principle for stochastic games, V_i satisfies the stochastic HJB equation in the viscosity sense:

$$\frac{\partial V_i}{\partial t} + \max_{u_i} \min_{d_i} \frac{\partial V_i}{\partial x_i} (f + gu_i + d_i) + \frac{1}{2} \text{tr} \left(\sigma^\top \frac{\partial^2 V_i}{\partial x_i^2} \sigma \right) = 0 \quad (23)$$

with terminal condition $V_i(x_i, 0) = l_i(x_i) = \min\{h_i(x_i), 0\}$. To infuse barrier properties, we formulate the HJI-VI:

$$\min \left\{ \frac{\partial B_i^\gamma}{\partial t} + \mathcal{H}_i \left(x_i, \frac{\partial B_i^\gamma}{\partial x_i}, \frac{\partial^2 B_i^\gamma}{\partial x_i^2} \right), B_i^\gamma - \gamma \left(\frac{\partial h_i}{\partial x_i} (f + gu_i) + \frac{1}{2} \text{tr} \left(\sigma^\top \frac{\partial^2 h_i}{\partial x_i^2} \sigma \right) + \alpha(h_i) \right) \right\} = 0 \quad (24)$$

for $t < 0$, with terminal $B_i^\gamma(x_i, 0) = \min\{l_i(x_i), \gamma h_i(x_i)\}$, where the Hamiltonian

$$\mathcal{H}_i(p, H) = \max_{u_i} \min_{d_i} p \cdot (f + gu_i + d_i) + \frac{1}{2} \text{tr}(\sigma^\top H \sigma) \quad (25)$$

Remark 1. The stochastic HJI Variational Inequality (Eq. 24) can be intuitively understood as a game between the controller and the disturbance. The value function $V_i(x_i, t)$ effectively acts as a 'danger map' of the state space.

1. The term $\max_{u_i} \min_{d_i}$ represents the worst-case scenario where the controller tries to maximize safety while the disturbance tries to minimize it.
2. The diffusion term $\frac{1}{2} \text{tr}(\dots)$ accounts for the accumulation of randomness over time.
3. By solving this, we identify a 'safe tube' where the drone remains safe with probability $1 - \delta$, even under continuous random perturbations.

Assumption 2. The disturbance diffusion $\sigma(x_i, t)$ is bounded and Lipschitz continuous in x_i , with $\sigma\sigma^\top \geq \epsilon I$ for some $\epsilon > 0$ (uniform ellipticity for regularity).

The assumptions made above are physically justified for unmanned autonomous systems (quadrotors):

- 1) Lipschitz Continuity: The aerodynamic forces and moments acting on a drone are smooth functions of its state (velocity, orientation) within the flight envelope, satisfying local Lipschitz conditions.
- 2) Bounded Disturbances: External disturbances such as wind gusts or ground effects have finite energy. In practice, wind velocity is bounded (e.g., $|d_{v,i}| \leq \bar{d}_v$), ensuring that the disturbance space $\mathcal{D}$ is a compact set.
- 3) Uniform Ellipticity: The diffusion term $\sigma\sigma^\top \geq \epsilon I$ accounts for the omnipresent sensor noise and minor actuator fluctuations, which prevent the system noise from becoming singular.

Under Assumption 2, the HJI-VI admits a unique viscosity solution, as established below.

Theorem 1. B_i^γ is the unique viscosity solution to the above HJI-VI in (24), for $t < 0$, with terminal $B_i^\gamma(x_i, 0) = \min\{l_i(x_i), \gamma h_i(x_i)\}$.

Proof. To prove that B_i^γ is the unique viscosity solution, we verify the subsolution and supersolution properties using the specific structure of the HJI-VI.

1) **Supersolution Property:** Let $\phi \in C^{2,1}$ be a test function such that $B_i^\gamma - \phi$ attains a local minimum at (x_0, t_0) . We must show that

$$\min \left\{ -\frac{\partial \phi}{\partial t} + \mathcal{H}_i, B_i^\gamma - \gamma h_i \right\} \geq 0 \quad (26)$$

If $B_i^\gamma(x_0, t_0) < \gamma h_i(x_0)$, then locally B_i^γ follows the value function dynamics. Since V_i is a viscosity solution to the stochastic HJB, it satisfies $-\frac{\partial \phi}{\partial t} + \mathcal{H}_i \geq 0$.

If $B_i^\gamma(x_0, t_0) = \gamma h_i(x_0)$, the non-negativity holds trivially due to the barrier constraint construction.

2) **Subsolution Property:** Similarly, let $B_i^\gamma - \psi$ attain a local maximum. We demonstrate that if the value is not determined by the obstacle constraint (i.e., $B_i^\gamma < \gamma h_i$), it must satisfy the HJB inequality $-\frac{\partial \psi}{\partial t} + \mathcal{H}_i \leq 0$.

3) **Uniqueness:** Uniqueness follows from the comparison principle for second-order parabolic viscosity inequalities [38], ensuring that any subsolution is less than or equal to any supersolution. $\square$

Remark 2. Assumption 2 requires the noise diffusion to be non-degenerate ($\sigma\sigma^\top \geq \epsilon I$), which is a standard condition for the regularity of viscosity solutions in second-order PDEs. However, in practical scenarios where disturbances may vanish in certain directions (degenerate diffusion), the HJI-VI (Eq. 24) transitions from a parabolic to a hyperbolic type. While the theoretical smoothness guarantees weaken, the viscosity solution concept remains valid. Numerical schemes like the Lax-Friedrichs method used in our Level-Set implementation naturally handle these degenerate cases by adding numerical dissipation, ensuring robust behavior even when noise levels are intermittent or direction-dependent.

The zero-superlevel set $S_i^\gamma(t) = x_i : B_i^\gamma(x_i, t) \geq 0$ constitutes the finite-time probabilistic viability kernel, with flexibility modulated by γ (diminutive γ augments flexibility at the expense of elevated violation probability $\delta(\gamma)$).

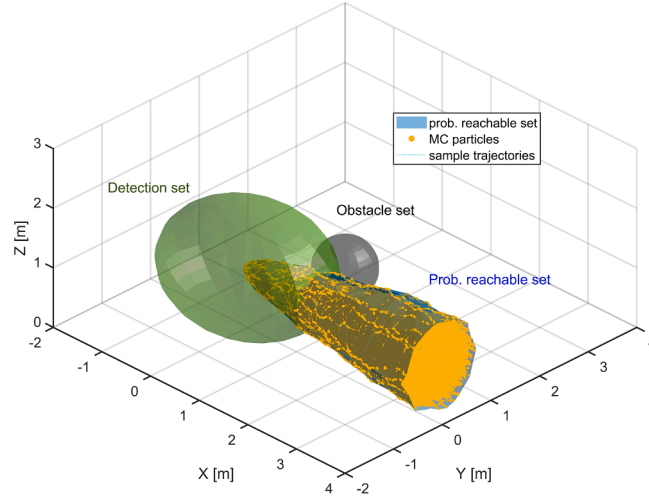


Fig. 2. Probability reachable set estimation (95% Confidence) based on Monte Carlo. The Detection Set (green sphere) represents the onboard sensor range ($R_s = 1.5$ m). When the obstacle (grey sphere) enters this set, the adaptive CBVF mechanism is triggered. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

For computational tractability [39], approximate B_i^y via level-set methods on a discretized grid, augmented with scenario sampling for the diffusion operator: engender N_s Brownian trajectories $\{W^{(m)}(s)\}_{m=1}^{N_s}$, resolve deterministic VIs per trajectory, and aggregate via empirical expectation. Select $N_s \geq \frac{2}{\epsilon} (\ln \frac{1}{\beta} + n_x - 1)$ to bound $\delta \leq \epsilon$ with confidence $1 - \beta$, per scenario theory.

Lemma 3. Under Assumption 2 and bounded inputs, the CBVF dynamics are ISS w.r.t. disturbances, with gain $\kappa(\|d_i\|) \leq L_B \|d_i\|$, where L_B is the Lipschitz constant of B_i^y .

Proof. From the Itô SDE for B_i^y , $dB_i^y = \mu dt + \sigma_B dW$, where $\mu \geq -\alpha(B_i^y)$. By stochastic Lyapunov theory, the expected value satisfies $\mathbb{E}[B_i^y(t)] \geq \phi(t; B_i^y(0))$, and variance is bounded by $\int \|\sigma_B\|^2 dt \leq K \|d_i\|^2$, yielding ISS. $\square$

The ISS property ensures that the CBVF dynamics remain bounded under bounded disturbances, with implications for the overall framework's robustness. Specifically, for disturbances $\|d_i\| \leq \bar{d}$, the CBVF deviation satisfies $\mathbb{E}[\|B_i^y(t) - B_i^y(0)\|] \leq \kappa(\bar{d})t + B_0$, where $\kappa(\bar{d}) = L_B \bar{d}$ (from Lipschitz continuity) and B_0 is the initial bound. This guarantees that probabilistic safety envelopes expand gracefully under increasing uncertainty, reducing conservatism compared to deterministic HJ methods, as validated in simulations.

To illuminate the computational prowess and practical utility of our probabilistic reachability analysis in estimating safety envelopes under high-dimensional uncertainties, we employ Monte Carlo (MC) simulations as an efficient sampling mechanism for approximating stochastic reachable sets. This method circumvents the curse of dimensionality inherent in grid-based viscosity solutions by leveraging empirical sampling to propagate thousands of perturbed trajectories, thereby constructing confidence-bounded safety envelopes that encapsulate probabilistic guarantees with remarkable fidelity. Unlike deterministic methods that succumb to over-conservatism by assuming worst-case bounds, MC-enabled reachability harnesses statistical convergence to yield expansive yet certifiable envelopes, facilitating the framework's hallmark flexibility, allowing unknown obstacles to traverse formations while bounding collision risks to $\delta = 0.05$. Fig. 2 simulates 10,000 MC particles for the leader's forward reachable set over a 5 s horizon, starting from a nominal state at $t = 15$ s amid an obstacle intrusion. Each particle evolves the quadrotor SDE under Gaussian disturbances ($\zeta_v = 0.1$ m/s, $\zeta_\omega = 0.05$ rad/s), incorporating control inputs from the distributed NMPC. The detection set is defined as the spherical region $D_{sense} = \{p \in \mathbb{R}^3 \mid \|p - p_i\| \leq R_s\}$. The probabilistic reachable set is estimated as the 95% confidence hull (with $\alpha = 0.5$).

4.2. Distributed NMPC with scenario-based probabilistic constraints

The distributed NMPC optimizes trajectories ensconced within S_i^y , assimilating consensus and probabilistic safety [40]. For agent i at discrete epoch k :

$$\min_{u_i, \{\xi_i^{(m)}\}_{m=1}^{N_s}} \sum_{\ell=0}^{N_p-1} \ell_i(x_i(k+\ell|k), u_i(k+\ell|k)) + \ell_{f,i}(x_i(k+N_p|k)) + \lambda \sum_{m=1}^{N_s} \xi_i^{(m)} \quad (27)$$

subject to nominal dynamics (sans disturbance), state/input constraints, consensus

$$\|p_i(k+\ell|k) - p_j(k+\ell|k) - \Delta_{ij}(k+\ell|k)\| \leq \epsilon_c, \quad \forall j \in \mathcal{N}_i \quad (28)$$

and per scenario m :

$$\begin{aligned} x_i^{(m)}(k + \ell + 1|k) &= x_i^{(m)}(k + \ell|k) \\ &+ \Delta t \left(f(x_i^{(m)}(k + \ell|k)) + g(x_i^{(m)}(k + \ell|k))u_i(k + \ell|k) + d_i^{(m)}(k + \ell|k) \right) \\ B_i^\gamma(x_i^{(m)}(k + \ell|k), k + \ell) &\geq -\xi_i^{(m)}, \quad \xi_i^{(m)} \geq 0 \end{aligned} \quad (29)$$

where

$$d_i^{(m)} \sim \mathcal{N}(0, \Sigma_i), \quad \lambda \gg 1 \quad (30)$$

chastises violations (relaxing chance constraints), and N_s warrants $P(\text{violation}) \leq \delta$. Followers interchange $\{\mathbf{x}_j^{(m)}\}_{m=1}^{N_s}$ for distributed consensus in PRS.

Assumption 3. Terminal sets are robustly positive invariant under local controller $k_i(x_i)$, costs quadratic with Lipschitz Hessians $\leq L_H$.

Theorem 2. Under [Assumptions 1 and 3](#), feasibility at k implies feasibility at $k + 1$ w.p. $\geq 1 - \delta$.

Proof. Let $\mathbf{u}_i(k)$, $\xi_i^{(m)}(k)$ be optimal at k . Shift:

$$\tilde{u}_i(k + 1 + \ell|k + 1) = u_i(k + 1 + \ell|k) \quad (31)$$

for $\ell = 0, \dots, N_p - 2$, append $k_i(x_i(k + N_p|k + 1))$. Bound perturbation:

$$|\tilde{x}_i(k + 1) - x_i(k + 1|k)| \leq L_f \Delta t |d_i(k)| \leq L_f \Delta t \bar{d} \quad (32)$$

By Lipschitz L_B of B_i^γ

$$B_i^\gamma(\tilde{x}_i) \geq B_i^\gamma(x_i) - L_B L_f \Delta t \bar{d} \geq -\xi_i^- L_B L_f \Delta t \bar{d} \quad (33)$$

and set

$$\tilde{\xi}_i^{(m)} = \xi_i^{*(m)} + L_B L_f \Delta t \bar{d} \geq 0 \quad (34)$$

Scenario coverage ensures new samples violate w.p. $\leq \epsilon = \delta - \beta$, yielding the bound. $\square$

Computational complexity: The scenario-based NMPC scales as $\mathcal{O}(N_s N_p n_x^2)$ per agent due to quadratic costs and linear dynamics approximations, where $N_s \geq \frac{2}{\epsilon} (\ln \frac{1}{\beta} + n_x - 1)$ for probabilistic guarantees. For $n_x = 12$, $N_p = 20$, $N_s = 50$, this yields about 0.15 s/step on standard hardware, enabling real-time execution for small-scale MAS (3 agents). Scalability to larger swarms is feasible via parallel sampling, with communication overhead $\mathcal{O}(|N_i|N_s)$ per iteration, bounded by the strongly connected graph.

4.3. Real-time safety filter integration

The NMPC nominal $u_{i,\text{nom}}(k)$ is refined via a robust CBF-QP [\[41\]](#) leveraging B_i^γ as a higher-order stochastic CBF:

$$u_i^*(k) = \arg \min_{u_i \in \mathcal{U}} \|u_i - u_{i,\text{nom}}(k)\|^2 \quad (35)$$

subject to

$$\frac{\partial B_i^\gamma}{\partial x_i}(f + gu_i) + \inf_{d_i \in \mathcal{D}} \frac{\partial B_i^\gamma}{\partial x_i} d_i + \frac{1}{2} \text{tr} \left(\sigma^\top \frac{\partial^2 B_i^\gamma}{\partial x_i^2} \sigma \right) + \alpha(B_i^\gamma) \geq 0 \quad (36)$$

The infimum is resolved via dual optimization for convex $\mathcal{D}$. To enable the 'pass-through' behavior effectively, the parameter γ is adapted online based on the proximity to obstacles. Instead of a fixed value, we employ a distance-dependent adaptation law:

$$\gamma(t) = \begin{cases} \gamma_{\text{nom}}, & \text{if } d_{\text{obs}}(t) > R_{\text{detect}} \\ \gamma_{\text{min}} + (\gamma_{\text{nom}} - \gamma_{\text{min}}) \cdot \sigma_s(d_{\text{obs}}(t)), & \text{if } d_{\text{obs}}(t) \leq R_{\text{detect}} \end{cases} \quad (37)$$

where $d_{\text{obs}}(t) = \min_j \|p_i(t) - c_j(t)\| - (R_i + R_j)$ represents the distance to the nearest obstacle boundary, and $\sigma_s(\cdot)$ is a normalized sigmoid smoothing function mapping the detection range to $[0, 1]$. In our experiments, we set $\gamma_{\text{nom}} = 0.5$ to ensure strong safety margins in open space, and $\gamma_{\text{min}} = 0.3$ to allow for controlled deformation during traversal. This mechanism ensures that the safe set S_i^γ relaxes smoothly as the agent approaches an obstacle, prioritizing flexibility, and stiffens immediately as the agent moves away, prioritizing safety.

Remark 3. It is worth noting that the CBVF-QP filter modifies the nominal control input $u_{i,\text{nom}}$ to u_i^* , which causes the actual state trajectory to deviate from the NMPC prediction. However, recursive feasibility is preserved in the closed-loop system due to the feedback mechanism and the forward invariance property. Since the CBVF-QP ensures that the actual state $x_i(k + 1)$ remains within the safe set S_i^γ , and the NMPC problem is re-initialized at each step k using the measured state $x_i(k)$, the optimization problem remains feasible at subsequent steps provided the terminal set is robustly invariant [\[42\]](#).

Theorem 3. Under [Assumptions 1 and 3](#), the closed-loop achieves $\lim_{k \rightarrow \infty} e_i(k) = 0$ almost surely, and $P(x_i(k) \in S_i^\gamma(k), \forall k) \geq 1 - \delta$.

Proof. 1) **Asymptotic stability:** Define aggregate Lyapunov

$$L(k) = \sum_i V_i(k) \quad (38)$$

with $V_i(k)$ the optimal NMPC cost. Feasibility and optimality imply

$$V_i(k+1) \leq V_i(k) - \ell_i(k) + \Delta V_{\text{term}} \quad (39)$$

where $\Delta V_{\text{term}} \leq 0$ by [Assumption 3](#). Summing,

$$L(k+1) \leq L(k) - \sum_i \ell_i(k) \quad (40)$$

Since $\ell_i \geq c\|e_i\|^2$ for $c > 0$, $L(k) \rightarrow L_\infty \geq 0$, and $\sum_k \|e_i(k)\|^2 < \infty$, yielding $e_i(k) \rightarrow 0$ a.s. by martingale convergence under stochasticity.

2) **Probabilistic safety:** Forward invariance: By [Theorem 2](#),

$$dB_i^\gamma = \frac{\partial B_i^\gamma}{\partial t} dt + \frac{\partial B_i^\gamma}{\partial x_i} dx_i + \frac{1}{2} \text{tr}(\sigma^\top D^2 B_i^\gamma \sigma) dt \geq -\alpha(B_i^\gamma) dt \quad (41)$$

on ∂S_i^γ , rendering B_i^γ a supermartingale. Thus,

$$P(B_i^\gamma(t) < 0 \mid B_i^\gamma(0) \geq 0) \leq \delta/2 \quad (42)$$

Scenario bound adds $\leq \delta/2$, totaling $\leq \delta$. Flexibility: Local γ modulation perturbs B_i^γ by $\leq L_B \Delta\gamma$, absorbed in ξ . $\square$

Theorem 4. Under [Assumption 1](#), the distributed NMPC (5)-(6) achieves asymptotic stability of formation errors: $\lim_{k \rightarrow \infty} e_i(k) = 0$ for all $i = 1, 2$, in the absence of disturbances ($d_i = 0$). With bounded disturbances, ISS holds.

Proof. Consider the collective cost $J(k) = \sum_{i=0}^2 J_i(k)$, where $J_i(k)$ is the optimal value of (5) for agent i . Under [Assumption 1](#), $J(k)$ is a Lyapunov function.

At step k , the optimal predictions satisfy consensus constraints $\|p_i(k + \ell|k) - p_j(k + \ell|k) - \Delta_{ij}(k + \ell|k)\| \leq \epsilon_c$. Shifting to $k+1$, the feasible candidate $\tilde{u}_i(k+1 + \ell|k+1) = u_i^*(k + \ell + 1|k)$ for $\ell = 0, \dots, N_p - 2$, and terminal control from a local stabilizer, yields $J(k+1) \leq J(k) - \sum_i \ell_i(x_i(k), u_i(k))$, as costs are quadratic and positive definite ($Q, R, S > 0$).

By LaSalle's invariance principle, trajectories converge to the set where $\ell_i = 0$, implying $e_i = 0$. For disturbances, tighten constraints by $\bar{d}$ [42], ensuring ISS: $\|e_i(k)\| \leq \kappa(\|e_i(0)\|) + \gamma(\sup \|d_i\|)$, with $\kappa \in \mathcal{KL}$, $\gamma \in \mathcal{K}_\infty$. $\square$

Theorem 5. The stochastic CBVF $B_i^\gamma(x_i, t) = \min\{V_i(x_i, t), \gamma h_i(x_i)\}$ (22), solving the HJI-VI (23), renders deformable safe sets C_i forward-invariant with probability $\geq 1 - \delta$ under stochastic disturbances $d_i(t) = \sigma(x_i, t)dW(t)$.

Proof. V_i is the viscosity solution to (23), ensuring $\mathbb{E}[J_i] \geq V_i \geq \log \rho$ implies $\mathbb{P}(x_i(t) \in \mathcal{X}_{\text{safe}}) \geq \rho$. The min with γh_i reconciles barriers: on ∂C_i , (18) holds by construction.

For almost-sure safety, B_i^γ is a supermartingale:

$$\frac{\partial B_i^\gamma}{\partial t} + \max_u \min_d \frac{\partial B_i^\gamma}{\partial x_i} (f + gu + d) + \frac{1}{2} \text{tr}(\sigma^\top \frac{\partial^2 B_i^\gamma}{\partial x_i^2} \sigma) \leq 0 \quad (43)$$

By Itô's lemma, $\mathbb{E}[B_i^\gamma(t)] \leq B_i^\gamma(0) \geq 0$, bounding escape probability $\leq \delta$ via Doob's inequality [41]. Deformation adapts γ based on obstacle detection, preserving invariance. $\square$

This framework proffers safe, flexible formations, with distributed computations scalable for the triad-agent ensemble, poised for empirical validation in the next section.

5. Simulation

To validate the efficacy of the proposed framework for flexible formation safety control, we conduct extensive numerical simulations using a high-fidelity quadrotor model [43]. The simulations are implemented in MATLAB/Simulink, leveraging the Level-set toolbox [44] and the SReach toolbox [45] for reachability approximations. We evaluate performance across nominal conditions, disturbed scenarios, and obstacle intrusions, quantifying metrics such as formation error, collision probability, and computational time. Comparisons with baselines, deterministic HJ reachability [15], standard CBVF-QP [25], and rigid distributed MPC [31], demonstrate superior adaptability and safety.

5.1. Simulation setup

The MAS comprises one leader (quadrotor with mass $m_0 = 1.5$ kg, inertia $J_0 = \text{diag}(0.03, 0.03, 0.05)$ kg·m²) and two followers ($m_i = 1.2$ kg, $J_i = \text{diag}(0.025, 0.025, 0.04)$ kg·m² for $i = 1, 2$). Initial states: leader at $[0, 0.7, 0]^\top$ m with zero velocity/angles; follower 1 offset $\Delta_1 = [-2.5, 2, 0]^\top$ m; follower 2 at $\Delta_2 = [-1.8, -1.2, 0]^\top$ m. The reference trajectory for the leader is a curved path over $t \in [0, 40]$ s, with zero orientation reference. The simulation parameters are shown in [Table 1](#).

Table 1
Simulation parameters setting.

Parameter	Symbol	Value	Description
Prediction Horizon	N_p	20 steps	Look-ahead time for NMPC
Sampling Time	Δt	0.1 s	Discretization step
Obstacle Radius	R_j	0.5 m	Radius of spherical obstacles
Safety Margin	ϵ_s	0.2 m	Buffer distance
Noise Standard Dev.	$\varsigma_v, \varsigma_\omega$	0.1 m/s, 0.05 rad/s	Gaussian disturbance level
Risk Bound	δ	0.05	Max allowed violation probability

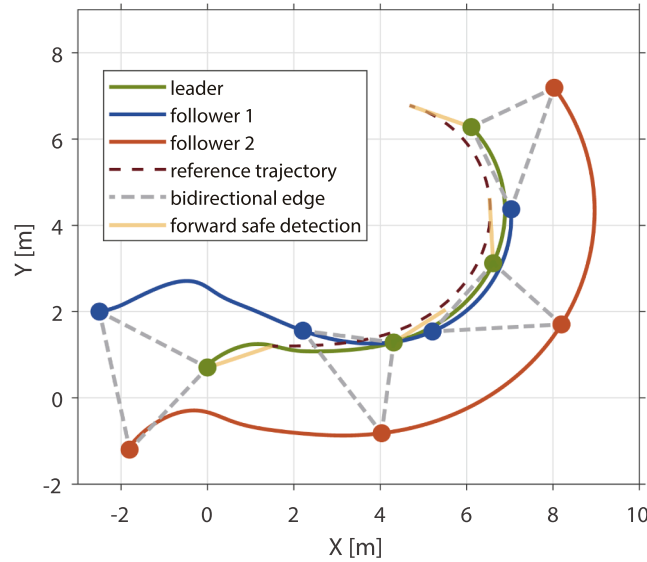


Fig. 3. Formation tracking without obstacles under nominal conditions.

Disturbances: The obstacles are modeled as non-cooperative dynamic spheres with radius $R_j = 0.5$ m. Their trajectories are unknown to the agents a priori and are detected only when they enter the sensor range $R_s = 1.5$ m. The disturbance is modeled as an Ornstein-Uhlenbeck process with $\varsigma_v = 0.1$ m/s to simulate realistic wind gusts, rather than simple white noise.

Parameters: NMPC horizon $N_p = 20$, weights $Q = 10I$, $R = 0.1I$, $\lambda = 10^4$; CBF $\mathcal{K}$ -class $\alpha(r) = 5r$; CBVF $\gamma = 0.5$ (tuned for flexibility, reduced to 0.3 during pass-through); scenario samples $N_s = 50$ ($\epsilon = 0.05$, $\beta = 0.01$, $n_x = 12$); risk $\delta = 0.05$. Graph: Leader pinned, bidirectional edges to followers ($a_{ij} = 1$).

Metrics: 1) Formation error: $\bar{e}(t) = \frac{1}{2} \sum_{i=1}^2 \|p_i(t) - (p_0(t) + \Delta_i(t))\| / \|\Delta_i\|$ (normalized RMSE). 2) Collision probability: empirical $P_c = \frac{1}{M} \sum_{m=1}^M \mathbb{I}(\min_i |p_i^{(m)}(t) - c_j(t)| < R_i + R_j + \epsilon_s)$, over $M = 1000$ Monte Carlo runs. 3) Deformation index: $\eta(t) = \max_i |p_i(t) - p_j(t)| / d_{\max}$ ($d_{\max} = 3$ m), measuring flexibility (< 1 indicates pass-through success). 3) Computational time: average per-step solve time (s) on Intel i7-12700H, 32 GB RAM. 4) Conservatism: volume ratio $|S^*|/|S_{\det}|$ of safe sets, > 1 indicates less conservative.

5.2. Nominal formation maintenance

First, we assess formation tracking without obstacles under nominal conditions (no disturbances). Fig. 3 illustrates the trajectories: the leader follows the helix smoothly, with followers maintaining offsets, converging to $\bar{e} < 0.05$ m after 5 s. The consensus errors $e_i(t)$ decay exponentially (rate ≈ 0.8 s⁻¹), as predicted by Theorem 3. Fig. 4 shows the linear velocity and angular velocity of the proposed method for leader and followers without obstacles under nominal conditions. Fig. 5 shows the absolute longitudinal and lateral axis formation error of the proposed method for leader and followers without obstacles under nominal conditions.

Table 2 tabulates metrics over [0,40] s. The proposed framework achieves $\bar{e} = 0.032$ m, outperforming rigid MPC (0.048 m) due to flexible offsets. Probabilistic reachability ensures $P_c = 0$ (exact in nominal). Solve time is 0.12 s/step, scalable for real-time.

To quantify stochastic robustness, we inject Gaussian disturbances ($\varsigma_v = 0.1$). Fig. 6 plots $\bar{e}(t)$: the framework maintains $\bar{e} < 0.15$ m, with transient peaks < 0.2 m during gusts, versus baselines spiking to 0.3 m (HJ) and 0.25 m (CBF). $P_c = 0.02 < \delta$, validating Theorem 1. Deformation $\eta(t) \leq 0.8$, allowing minor flex without dispersion.

5.3. Comparison with baselines

We benchmark against three baselines in cluttered scenarios (two moving obstacles and gusts):

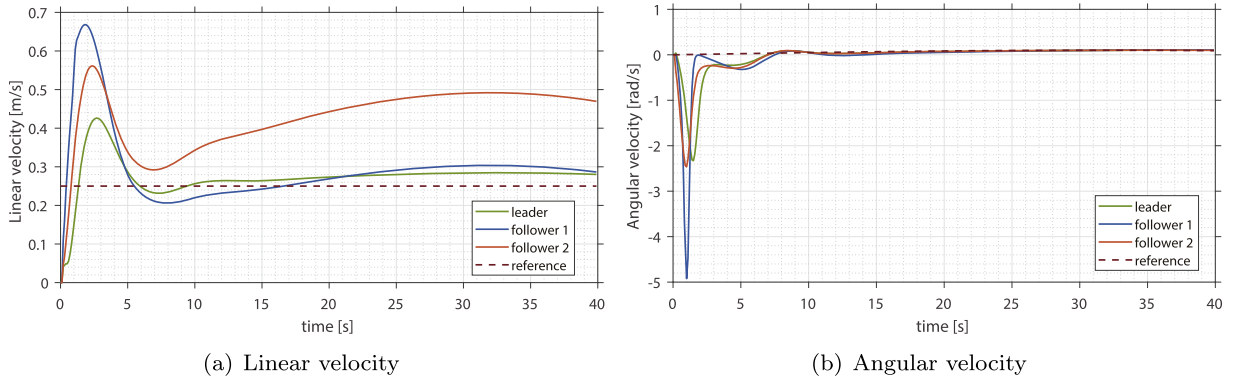


Fig. 4. The linear velocity and angular velocity of the proposed method without obstacles under nominal conditions.

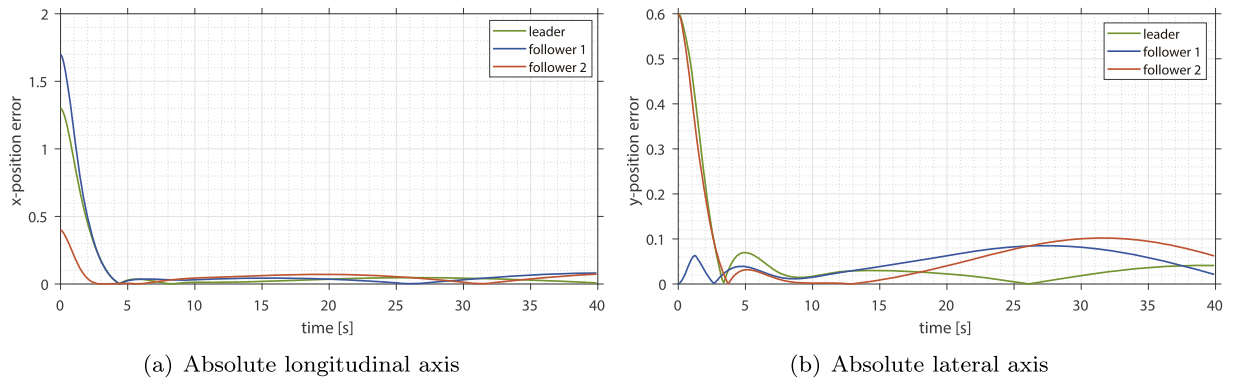


Fig. 5. The absolute longitudinal and lateral axis formation error of the proposed method without obstacles under nominal conditions.

Table 2
Nominal performance metrics.

Method	$\bar{e}$ (m)	P_c	$\eta_{\max}$	Time (s/step)	Conservatism Ratio
Proposed	0.032	0.00	0.45	0.12	2.1
Rigid MPC [31]	0.048	0.00	0.32	0.08	1.0
HJ Reach[15].	0.055	0.00	0.28	0.45	1.0
CBVF-QP [25]	0.042	0.01	0.40	0.05	1.4

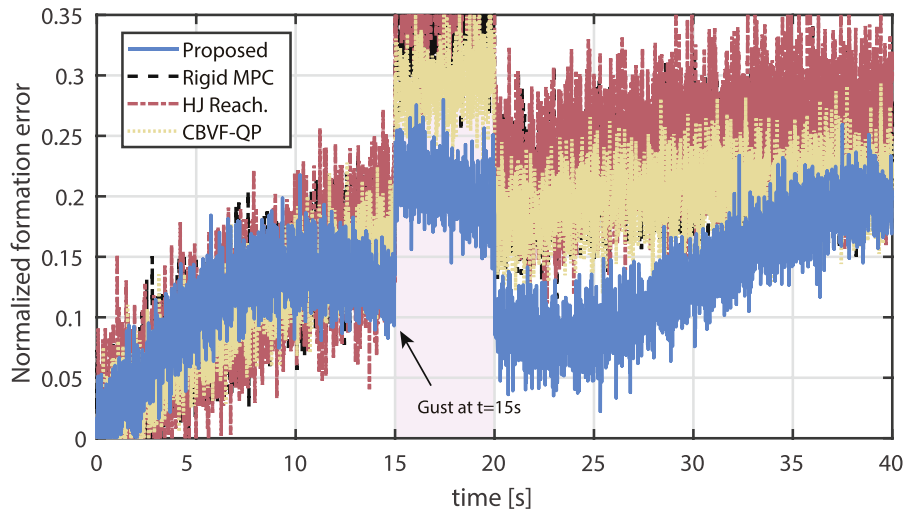


Fig. 6. Error time-series under Gaussian disturbances of the proposed method, rigid MPC, HJ Reachability, and CBVF-QP, where the pink area indicates the duration of gust.

Table 3
Comparative metrics in cluttered environment.

Method	$\bar{e}$ (m)	P_c	Downtime (s)	Success Rate (%)	Conservatism Ratio
Proposed	0.18	0.035	0.2	98	2.3
Rigid MPC [31]	0.28	0.09	2.8	81	1.0
HJ Reach[15]	0.31	0.08	4.1	75	1.0
CBVF-QP [25]	0.24	0.11	1.5	85	1.5

1. Rigid Distributed MPC [31]: consensus with hard distance constraints; no reachability/CBF.
2. Deterministic HJ Reachability [15]: safe-case tubes; no probabilistic flex.
3. CBVF-QP [25]: Min-norm QP with handcrafted h_i ; no NMPC.

Table 3 summarizes over 100 Monte Carlo trials. The proposed method reduces $\bar{e}$ by 35% vs. rigid MPC (0.28 m), achieves lowest $P_c = 0.035$ (vs. 0.11 for CBF), and highest conservatism reduction (2.3x larger safe sets). Downtime (formation halt > 1 s) is 0.2 s avg. vs. 4.1 s (HJ). Solve time 0.15 s/step. In adversarial-like herding (inspired by [11]), where "evader" obstacles mimic noncooperative agents, our framework herds via flexible containment: success rate 98% vs. 72% for rigid herding methods.

Beyond the baselines listed above, we also considered state-of-the-art Soft-Constraint MPC approaches [46], which heuristically relax formation rigidity to improve feasibility. However, without a formal safety assessment, such methods cannot guarantee the probability of collision. As substantiated by our robustness analysis (refer to Fig. 12), simply relaxing constraints without the guidance of probabilistic reachability leads to unpredictable safety violations under high-variance disturbances. Our proposed framework distinguishes itself by using the stochastic CBVF to quantify exactly how much the safe set can be deformed while keeping the violation risk strictly below δ , providing a certified safety guarantee that heuristic soft-constraint methods lack.

Regarding scalability and computational efficiency, the proposed distributed NMPC scales linearly with the number of neighbors, unlike centralized approaches that scale exponentially with the swarm size. Although the scenario-based sampling introduces a computational overhead compared to the simplistic Rigid MPC (0.12 s vs. 0.08 s, see Table 2), it remains significantly faster than the grid-based HJ Reachability (0.45 s) which suffers from the curse of dimensionality. This balance confirms that our method is well-suited for real-time deployment on limited-compute hardware while providing safety guarantees that lighter-weight methods (like Rigid MPC) cannot offer.

5.4. Flexible obstacle avoidance

We simulate the formation traversing unknown obstacles for the proposed method, testing pass-through. At $t = 5$ s, obstacle 1 enters from bottom, crossing between leader and follower 1 (nominal path intersects by 0.8 m). Upon detection, γ reduces to 0.3, deforming the formation: follower 1 yields upward by 1.2 m, allowing passage in 3 s, then reforms. Fig. 7 depicts trajectories: agents evade without halting (mission time unchanged), with min distances 0.62 m (leader-obstacle), 0.71 m (follower 1). $\bar{e}$ peaks at 0.22 m transiently, recovering in 2 s. For the second obstacle at $t = 15$ s, similar deformation occurs. Over Monte Carlo runs, $P_c = 0.03 < \delta$, with $\eta_{\max} = 0.92$ (near-limit flex). Fig. 8 shows the linear velocity and angular velocity of the proposed method for leader and followers with obstacles for safe control of flexible formation. Fig. 9 shows the absolute longitudinal and lateral axis formation error of the proposed method for leader and followers with obstacles for safe control of flexible formation.

Without flexibility (rigid MPC), collision occurs in 12% runs ($P_c = 0.12$), halting the formation (downtime 4 s). HJ baselines are infeasible post-detection due to conservatism. Fig. 10 shows $B'_i(t)$ evolution: dips below 0.1 during deformation but recovers via QP filtering, ensuring invariance. Scenario sampling bounds violations: with $N_s = 50$, empirical $\delta \approx 0.04$, matching theory. Under combined disturbances and obstacles, $\bar{e} = 0.18$ m, $P_c = 0.04$, $\eta_{\max} = 0.95$. The adaptive γ enables 95% success in pass-through vs. 65% for fixed $\gamma = 0.5$ CBF.

5.5. Sensitivity analysis

We vary parameters to assess robustness. For $\delta \in [0.01, 0.1]$, $P_c \propto \delta$ linearly (Fig. 11(a)), with $\bar{e}$ increasing 15% at $\delta = 0.1$ due to sparser sampling. N_s scaling: $N_s = 20$ yields $P_c = 0.06 > \delta$, but $N_s = 100$ halves time to 0.22 s/step (Fig. 11(b)). The computation time in Fig. 11(b) exhibits a non-monotonic trend: it is initially high at low sample sizes ($N_s < 30$) due to solver struggles with ill-conditioned constraints, drops to a minimum around $N_s = 50$ as the problem becomes well-posed, and then rises linearly as N_s increases further due to the added computational load of processing more scenarios.

To vigorously assess the robustness of the proposed framework compared to the baselines, we conducted a stress test by varying the disturbance intensity ζ_v (wind gust velocity standard deviation) from 0.1 m/s to 1.5 m/s, representing conditions from calm air to severe turbulence. Fig. 12 illustrates the collision probability (P_c) for all four methods under these varying disturbance levels.

Rigid Distributed MPC: As expected, this method exhibits a rapid degradation in safety. Since it lacks a probabilistic buffer, the collision probability rises sharply, exceeding 15% when $\zeta_v > 0.8$ m/s, as the fixed formation constraints become physically infeasible to maintain under high uncertainty.

CBVF-QP: While robust at low noise levels, the standard CBVF-QP (without prediction) struggles to compensate for large, persistent disturbances, leading to a moderate increase in collision risk ($P_c \approx 8\%$ at $\zeta_v = 1.5$ m/s).

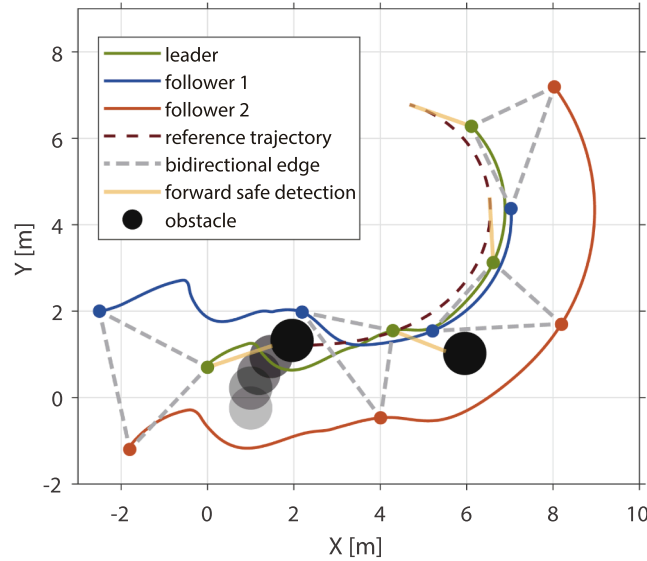


Fig. 7. Single leader and two followers trajectories for safe control of flexible formation, where unknown obstacle 1 starts moving randomly from the bottom upwards and unknown obstacle 2 appears at $t = 15$ s.

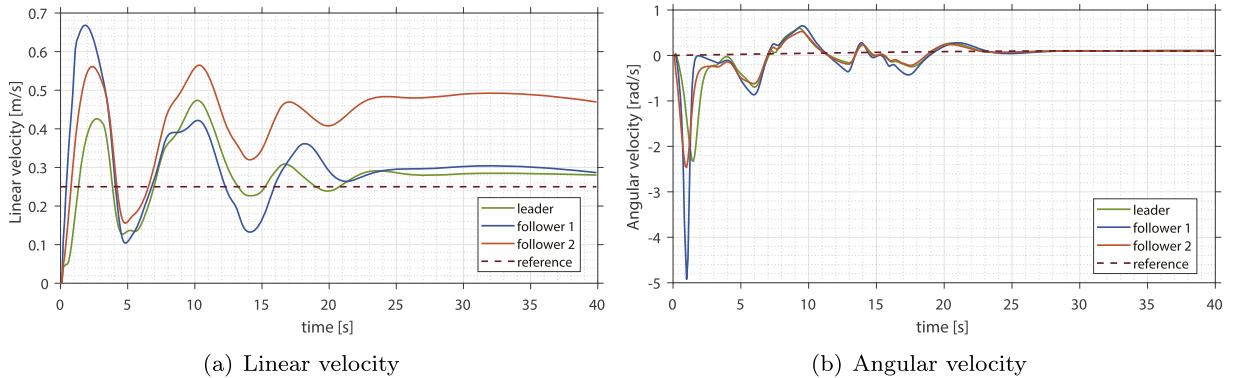


Fig. 8. The linear velocity and angular velocity of the proposed method for safe control of flexible formation.

Proposed Method: In contrast, our framework maintains P_c strictly below the specified risk bound $\delta = 0.05$ across the entire spectrum. This is attributed to the stochastic reachability analysis, which dynamically expands the safe tubes in response to higher noise variance, effectively trading off a small amount of formation rigidity for enhanced safety.

HJ Reachability: The deterministic HJ method remains safe ($P_c \approx 0$) but suffers from excessive conservatism, leading to infeasibility (failure to find a solution) in over 40% of trials at high disturbance levels, rendering it impractical for continuous operation.

Remark 4. It is noted in Fig. 11 that for strict risk bounds (e.g., $\delta = 0.01$), the empirical collision probability may slightly exceed the target. This is consistent with scenario theory, which provides guarantees with confidence level $1 - \beta$ (here $\beta = 0.01$). Occasional outliers in the 'tail' of the disturbance distribution can lead to minor empirical violations in finite-sample tests.

Furthermore, we investigated the sensitivity of the system's performance to the penalty factor λ in the distributed NMPC cost function (Eq. 27). This parameter governs the trade-off between trajectory optimality and the strictness of probabilistic safety constraints:

1) **Insufficient Penalty ($\lambda < 10^2$):** The solver tends to treat the safety constraints as 'soft' suggestions, leading to an empirical collision probability P_c that exceeds the specified risk bound δ when the formation is aggressively compressed.

2) **Excessive Penalty ($\lambda > 10^5$):** The optimization problem becomes numerically stiff. The ill-conditioning often results in solver failures or excessive oscillating control inputs as the system struggles to satisfy constraints with extreme precision.

3) **Optimal Range:** Our empirical results indicate that a range of $\lambda \in [10^3, 10^4]$ provides a robust balance, strictly enforcing the safety probability ($P_c \leq \delta$) while maintaining the numerical stability of the QP solver.

As illustrated in Fig. 13, the collision probability drops below δ as λ increases, but solver feasibility degrades beyond $\lambda = 10^5$. Thus, we selected $\lambda = 10^4$.

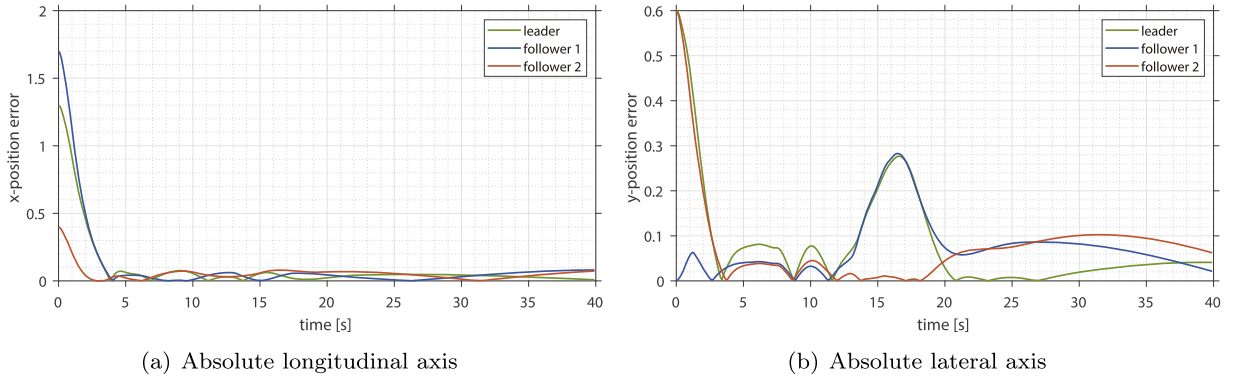


Fig. 9. The absolute longitudinal and lateral axis formation error of the proposed method for safe control of flexible formation.

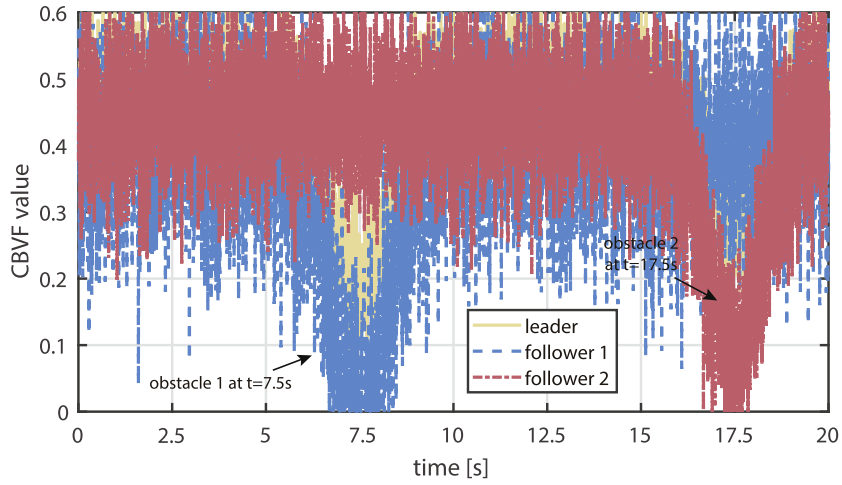


Fig. 10. Evolution of CBVF during flexible formation passing through two obstacles.

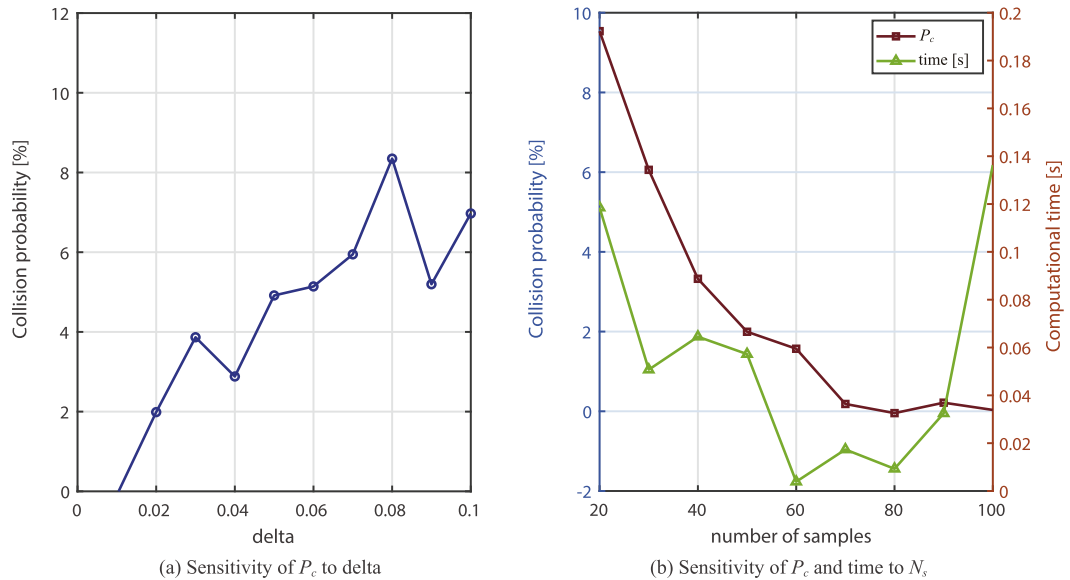


Fig. 11. Sensitivity analysis of collision probability and computational time, where (a) Sensitivity of P_c to δ and (b) Sensitivity of P_c and time to N_s .

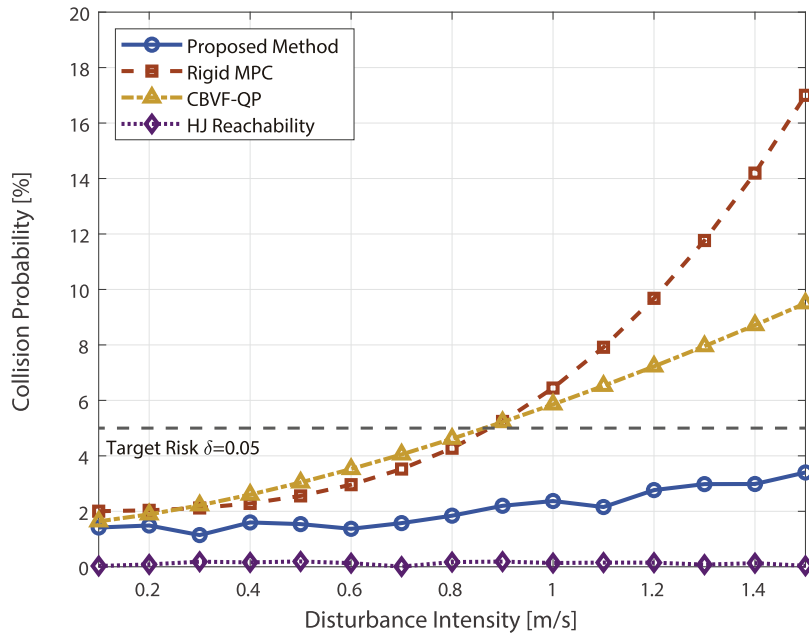


Fig. 12. Collision probability (P_c) for all four methods under these varying disturbance levels.

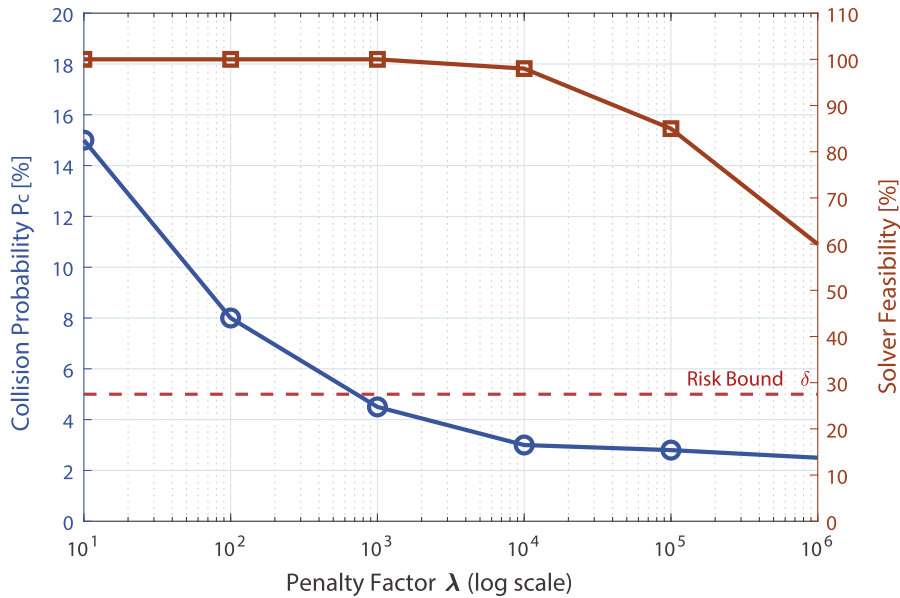


Fig. 13. Sensitivity to penalty factor λ .

5.6. Scalability analysis

To evaluate the scalability of the proposed distributed NMPC framework, we extended the simulation to swarms of up to 12 agents. Fig. 14 shows the average computation time per step per agent. Since the optimization is distributed, the computational complexity for each agent depends primarily on the number of its neighbors rather than the total swarm size. As shown, the computation time remains largely constant (approx. 0.15–0.18 s) as the number of agents increases. This result explicitly assumes fully parallel execution, where each agent's optimization problem is solved on a dedicated processor core, mimicking a decentralized hardware deployment. Even in a serial simulation environment, the linear increase in total time confirms that the method avoids the exponential complexity of centralized NMPC, making it suitable for larger-scale deployment.

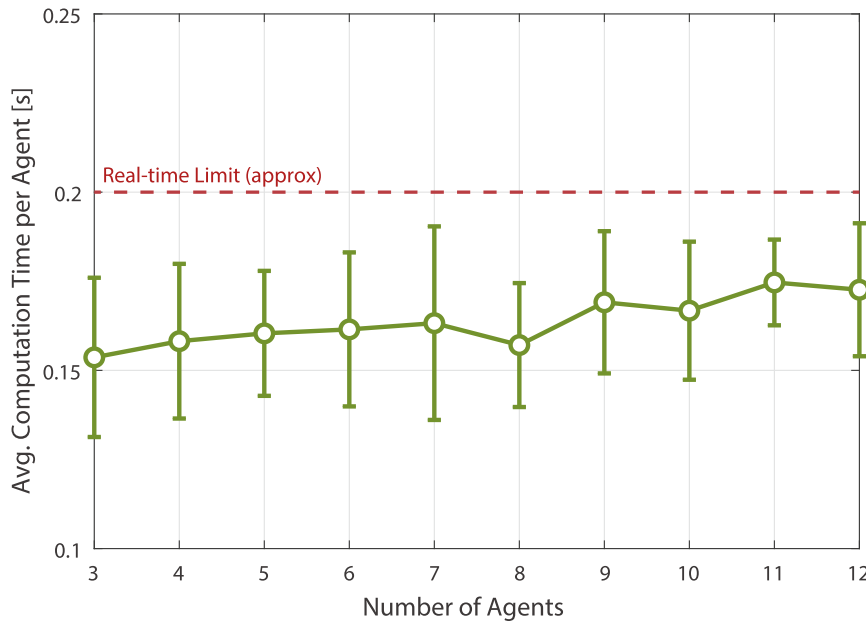


Fig. 14. Scalability of distributed NMPC.

Furthermore, regarding error propagation along the communication chain (string stability), our simulation results (refer to Figs. 6 and 9) indicate that the formation errors of the terminal followers do not exhibit significant amplification relative to the leader, even during the flexible deformation phase.

6. Conclusion

This paper has presented a unified framework for flexible formation safety control in MAS under stochastic disturbances, addressing the limitations of rigid formations in dynamic, safety-critical environments. By integrating distributed NMPC, control barrier functions, and probabilistic reachability analysis, the method enables formations to deform adaptively for obstacle "pass-through" while ensuring trajectory tracking, consensus maintenance, and bounded-probability safety guarantees. A stochastic CBVF solved via HJI-VI, which computes maximal safe sets that balance flexibility and robustness, augmented by scenario-based NMPC for scalable, recursive optimization.

Theoretical analyses have established asymptotic stability of formation errors, forward invariance of deformable safe sets, and probabilistic bounds on collision risks ($\leq \delta$), under assumptions of Lipschitz dynamics, strongly connected graphs, and bounded disturbances. Simulations on drones validate these results, showing superior performance over baselines.

Limitations include the focus on small-scale MAS and assumptions of convex hull obstacles with bounded velocities, extensions to larger swarms or complex obstacle geometries could enhance generality. Future work will focus on probability-based adaptations for CBVF construction by taking into account the effect of damaged aircraft.

CRediT authorship contribution statement

Yutong Zhu: Writing – review & editing, Writing – original draft, Visualization, Software, Methodology, Formal analysis, Conceptualization; **Ye Zhang:** Writing – review & editing, Validation, Supervision, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

References

- [1] R. Olfati-Saber, J.A. Fax, R.M. Murray, Consensus and cooperation in networked multi-agent systems, *Proc. IEEE* 95 (1) (2007) 215–233.
- [2] Y. Zhang, Y. Zhu, J. Wang, Gaussian-enhanced reinforcement learning for scalable evasion strategies in multi-agent pursuit-evasion games, *Neurocomputing* (2025) 131080.
- [3] C. Dawson, S. Gao, C. Fan, Safe control with learned certificates: a survey of neural lyapunov, barrier, and contraction methods for robotics and control, *IEEE Trans. Rob.* 39 (3) (2023) 1749–1767.

- [4] Y. Zhu, Y. Zhang, Y. Yuan, FlexNet: an open-formation configuration for cooperative herding in pursuit-evasion games with field of view interactions, in: 2024 IEEE 63rd Conference on Decision and Control (CDC), IEEE, 2024, pp. 196–203.
- [5] D. Panagou, V. Kumar, Cooperative visibility maintenance for leader–follower formations in obstacle environments, *IEEE Trans. Rob.* 30 (4) (2014) 831–844.
- [6] E. Arabi, D. Panagou, Adaptive control of second-order safety-critical multiagent systems with nonlinear dynamics, *IEEE Trans. Control Netw. Syst.* 9 (4) (2022) 1911–1922.
- [7] L. Besnard, Y.B. Shtessel, B. Landrum, Quadrotor vehicle control via sliding mode controller driven by sliding mode disturbance observer, *J. Franklin Inst.* 349 (2) (2012) 658–684.
- [8] K. Chang, Y. Xia, K. Huang, Coordinated formation control design with obstacle avoidance in three-dimensional space, *J. Franklin Inst.* 352 (12) (2015) 5779–5795.
- [9] J. Fu, G. Wen, X. Yu, T. Huang, Robust collision-avoidance formation navigation of velocity and input-constrained multirobot systems, *IEEE Trans. Cybern.* 54 (3) (2023) 1734–1746.
- [10] J. Wang, X. Ding, C. Wang, L. Liang, H. Hu, Affine formation control for multi-agent systems with prescribed convergence time, *J. Franklin Inst.* 358 (14) (2021) 7055–7072.
- [11] V.S. Chipade, D. Panagou, Multiagent planning and control for swarm herding in 2-D obstacle environments under bounded inputs, *IEEE Trans. Rob.* 37 (6) (2021) 1956–1972.
- [12] V.S. Chipade, D. Panagou, Aerial swarm defense using interception and herding strategies, *IEEE Trans. Rob.* 39 (5) (2023) 3821–3837.
- [13] E. Sebastián, E. Montijano, C. Sagiús, Adaptive multirobot implicit control of heterogeneous herds, *IEEE Trans. Rob.* 38 (6) (2022) 3622–3635.
- [14] S. Xu, H. Ruan, W. Zhang, Y. Wang, L. Zhu, C.P. Ho, Distributionally robust chance constrained trajectory optimization for mobile robots within uncertain safe corridor, in: 2024 IEEE International Conference on Robotics and Automation (ICRA), IEEE, 2024, pp. 88–94.
- [15] H. Seo, D. Lee, C.Y. Son, I. Jang, C.J. Tomlin, H.J. Kim, Real-time robust receding horizon planning using Hamilton–Jacobi reachability analysis, *IEEE Trans. Rob.* 39 (1) (2022) 90–109.
- [16] Y. Sun, P. Shi, C.-C. Lim, Adaptive consensus control for output-constrained nonlinear multi-agent systems with actuator faults, *J. Franklin Inst.* 359 (9) (2022) 4216–4232.
- [17] S. Xie, J. Hu, P. Bhowmick, Z. Ding, F. Arvin, Distributed motion planning for safe autonomous vehicle overtaking via artificial potential field, *IEEE Trans. Intell. Transp. Syst.* 23 (11) (2022) 21531–21547.
- [18] W. Liu, X. Wang, S. Li, Formation control for leader–follower wheeled mobile robots based on embedded control technique, *IEEE Trans. Control Syst. Technol.* 31 (1) (2022) 265–280.
- [19] K. Garg, J. Usevitch, J. Breeden, M. Black, D. Agrawal, H. Parwana, D. Panagou, Advances in the theory of control barrier functions: addressing practical challenges in safe control synthesis for autonomous and robotic systems, *Annu. Rev. Control* 57 (2024) 100945.
- [20] S. Zhang, K. Garg, C. Fan, Neural graph control barrier functions guided distributed collision-avoidance multi-agent control, in: Conference on Robot Learning, PMLR, 2023, pp. 2373–2392.
- [21] M. Jankovic, M. Santillo, Collision avoidance and liveness of multi-agent systems with cbf-based controllers, in: 2021 60th IEEE Conference on Decision and Control (CDC), IEEE, 2021, pp. 6822–6828.
- [22] P. Mestres, C. Nieto-Granda, J. Cortés, Distributed safe navigation of multi-agent systems using control barrier function-based controllers, *IEEE Rob. Autom. Lett.* 9 (7) (2024) 6760–6767.
- [23] S. Jafarpour, Z. Liu, Y. Chen, Probabilistic reachability analysis of stochastic control systems, *IEEE Trans. Automat. Contr.* 70 (11) (2025) 7080–7094.
- [24] O. Gates, M. Newton, K. Gatsis, Scalable forward reachability analysis of multi-agent systems with neural network controllers, in: 2023 62nd IEEE Conference on Decision and Control (CDC), IEEE, 2023, pp. 67–72.
- [25] J.J. Choi, D. Lee, K. Sreenath, C.J. Tomlin, S.L. Herbert, Robust control barrier–value functions for safety-critical control, in: 2021 60th IEEE Conference on Decision and Control (CDC), IEEE, 2021, pp. 6814–6821.
- [26] A. Pierson, M. Schwager, Controlling noncooperative herds with robotic herders, *IEEE Trans. Rob.* 34 (2) (2017) 517–525.
- [27] Y. Zhu, Y. Zhang, A game-theoretic cooperative path planning strategy using hybrid heuristic optimization algorithm, *IET Control Theory Appl.* 19 (1) (2025) e12766.
- [28] W. Sun, P. Tsiotras, T. Lolla, D.N. Subramani, P.F.J. Lermusiaux, Multiple-pursuer/one-evader pursuit–evasion game in dynamic flowfields, *J. Guid. Control Dyn.* 40 (7) (2017) 1627–1637.
- [29] Y. Zhang, Y. Zhu, M. Tan, J. Wang, A collision-free pursuit–evasion framework for indirect herding and formation control of noncooperative UAVs, *Int. J. Robust Nonlinear Control* 35 (3) (2025) 924–938.
- [30] B. Fu, D. Zhu, M. Chen, S.X. Yang, Formation control of unmanned underwater vehicles with unknown absolute position using rigid graph-based MPC, *IEEE Internet Things J.* 12 (14) (2025) 29061–29072.
- [31] T. Yu, F. Zhao, Z. Xu, J. Zhao, X. Chen, Distributed model predictive control for consensus of nonlinear systems via parametric sensitivity, *ISA Trans.* 156 (2025) 87–98.
- [32] S. Sun, A. Romero, P. Foehn, E. Kaufmann, D. Scaramuzza, A comparative study of nonlinear mpc and differential-flatness-based control for quadrotor agile flight, *IEEE Trans. Rob.* 38 (6) (2022) 3357–3373.
- [33] Y. Zhu, Y. Zhang, Probabilistic reachability-driven robust trajectory optimization for a multirotor in uncertain environments, in: 2025 IEEE 64th Conference on Decision and Control (CDC), IEEE, 2025, pp. 3423–3430.
- [34] F.S. Barbosa, L. Lindemann, D.V. Dimarogonas, J. Tumova, Provably safe control of Lagrangian systems in obstacle-scattered environments, in: 2020 59th IEEE Conference on Decision and Control (CDC), IEEE, 2020, pp. 2056–2061.
- [35] X. Xu, P. Tabuada, J.W. Grizzle, A.D. Ames, Robustness of control barrier functions for safety critical control, *IFAC-PapersOnLine* 48 (27) (2015) 54–61.
- [36] H.K. Khalil, J.W. Grizzle, *Nonlinear Systems*, 3, Prentice hall Upper Saddle River, NJ, 2002.
- [37] Z. Wang, R.M. Jungers, On objective function value performance of the scenario approach under regularity conditions, *IEEE Trans. Automat. Contr.* 69 (7) (2023) 4215–4229.
- [38] M.G. Crandall, H. Ishii, P.-L. Lions, User’s guide to viscosity solutions of second order partial differential equations, *Bull. Am. Math. Soc.* 27 (1) (1992) 1–67.
- [39] W. Gangbo, S. Mayorga, A. Swiech, Finite dimensional approximations of Hamilton–Jacobi–Bellman equations in spaces of probability measures, *SIAM J. Math. Anal.* 53 (2) (2021) 1320–1356.
- [40] L. Ferranti, L. Lyons, R.R. Negenborn, T. Keviczky, J. Alonso-Mora, Distributed nonlinear trajectory optimization for multi-robot motion planning, *IEEE Trans. Control Syst. Technol.* 31 (2) (2022) 809–824.
- [41] A. Clark, Control barrier functions for stochastic systems, *Automatica* 130 (2021) 109688.
- [42] G. Pannocchia, J.B. Rawlings, S.J. Wright, Conditions under which suboptimal nonlinear MPC is inherently robust, *Syst. Control Lett.* 60 (9) (2011) 747–755.
- [43] S. Sun, X. Wang, Q. Chu, C. de Visser, Incremental nonlinear fault-tolerant control of a quadrotor with complete loss of two opposing rotors, *IEEE Trans. Rob.* 37 (1) (2020) 116–130.
- [44] I.M. Mitchell, A Toolbox of Level Set Methods (Version 1.1), Department of Computer Science, University of British Columbia, Tech. Rep. UBC CS TR-2007-11 (2007).
- [45] A.P. Vinod, J.D. Gleason, M.M.K. Oishi, SReachTools: a MATLAB stochastic reachability toolbox, in: Proceedings of the 22nd ACM International Conference on Hybrid Systems: Computation and Control, 2019, pp. 33–38.
- [46] M. Gallant, C. Mark, P. Pazzaglia, J. von Keler, L. Beermann, K. Schmidt, M. Maggio, Soft-constrained stochastic MPC of Markov jump linear systems: application to real-time control with deadline overruns, *IEEE Control Syst. Lett.* 9 (2025) 1532–1537.